

Release Notes

Package REL1.0 — software 1.2.5

Document number	iW-ASCEJ-01-ReleaseNotes-R1.0-REL1.0
Revision	R1.0
Package release	REL1.0 (software 1.2.5)
Date	05-Oct-2026

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Release identification

Item	Value
Project	iW-ASCEJ-01 — iWave SDR platform, ZU47DR RFSoc
Package	iW-ASCEJ-01-DF-R1.0-REL1.0
Package release	REL1.0 (first formal release package)
Software version	1.2.5
Firmware PCIe ABI	4 (capabilities 0x7F)
Target device	xczu47dr-ffvg1517-2-i
Host	x86-64 Linux, Qt 5.15 / Qt 6, iwfg QDMA driver

Contents of the package

Folder	Delivered
HF	psu_init.c, psu_init.html (PS configuration), Vitis export.json, Hardware Interface Summary, PL hand-off list
FF	37 firmware sources; boot.bif, make_boot.sh, boot_probe.tcl, jtag_boot_test.tcl, linker fragment; 8 firmware documents and the register-header reference
BN	sdr_workstation (GUI), rfdc_ctl (CLI), with SHA-256 sums
SC	GUI sources (src/core, src/ui), backend helpers (c2h_stream, iwfg_h2c, UDP receiver), iwfg QDMA kernel driver, resources, CMake; rfdc_ctl + pcie_access
UM	User Manual (PDF), topic guides, developer notes and the full change log
AT	Acceptance Test Procedure, Acceptance Test Report, raw logs
TS	Host test suite (4 suites, 10 tests, 244 checks) and verification scripts
DF	API and engineering-calculation workbook (1,737 APIs, live calculator), these release notes, manifest

What is new in software 1.2.5

All 1.2.5 changes are in the GUI only. The board does not need to be reflashed, and the driver is unchanged.

Capture and transmit no longer break each other. The root cause is in the `iwfg` driver. On every C2H FIFO overflow it soft-resets the whole QDMA and rebuilds every queue, the H2C queue included, even while an H2C transfer is waiting on it. A 4-channel capture at 200 MSPS needs 3052 MiB/s, and the bench host drains about 2400 MiB/s, so this happens continuously. The result was tone disturbances, transmit stalls and captures that died or sat at 0.0 MB/s.

Change	Effect
Transmit guard	While the generator runs, a capture above 1600 MiB/s is reduced to 2 channels via PL GPIO mode. The GUI checks the result against the measured rate and restores the setting on stop.
Stalled-capture device reset	Silent capture for about 5 s → close transmit, then capture (the driver resets on the last close), then restart both. No dialog appears.
Ordered stop	Every stop or start leaves the device freshly reset
Same-size reopen	A stalled transmit helper is reopened at the same size before any step-down
Capture FIFO pipe	1 MiB (was 64 KiB)
QMC check	The ADC row warns when the ADC QMC gain reads 0

Earlier changes carried in this release (1.2.1 - 1.2.4)

Version	Change
1.2.4	H2C default buffer is 1 MiB (16 MiB never completed). Automatic step-down 1 MiB → 16 KiB on a stalled transfer. Capture helper restarts automatically. Stop/Start/close never block. Helper output goes to <code>sdr_backend.log</code> . Helper rows in the transmit chain.
1.2.3	H2C TRANSMIT CHAIN view read from the device. Verify loopback check.
1.2.2	Clean host transmit: coherent replay loop, write-once generator
1.2.1	DAC input source fixed: DDS and host were swapped end to end. The board now boots on the DDS.

The complete history (1.0.0 onward) is in `UM/02_Developer_Notes/CHANGELOG.md`.

Compatibility

Item	Note
Firmware	ABI 4 is required. A board on 1.2.0 firmware is still controlled correctly (tested), but should be reflashed.
pcie_regs.h	Must be identical in FF, SC/GUI and SC/host-tools (checked by audit_sync.sh)
Prebuilt binaries	glibc ≥ 2.38 (Ubuntu 24.04). Older systems: build from SC.
Backend helpers	Always built on the target host by the GUI build. Never shipped prebuilt.
Driver	Unchanged in this release. See the known limitations.

Verification

Check	Result
Package integrity, MANIFEST.sha256	PASS
Firmware ↔ GUI ↔ CLI contract (audit_sync.sh , 13 checks)	PASS
GUI build from SC (Qt 5.15)	PASS
TS-01 firmware DAC routing	17/17 PASS
TS-02/03 PCIe end-to-end, 1.2.1 and 1.2.0 firmware	17/17 + 17/17 PASS
TS-04 transmit path	20/20 PASS
TS-05 simulated loopback	13/13 PASS
TS-06 control window	39/39 PASS
TS-07 whole application, simulation	4/4 PASS
TS-08 real iwfg_h2c , emulated card	18/18 PASS
TS-09 send-and-receive chain with fault injection	73/73 PASS
TS-10 whole application, live session, emulated card	26/26 PASS

Known limitations

1. The driver's C2H overflow recovery resets H2C as well. The GUI works around it; the recommended driver fixes are documented in [UM/02_Developer_Notes/DRIVER_OVERFLOW_RECOVERY.md](#).
2. A 4-channel capture at 200 MSPS exceeds the measured host drain. It is not usable while transmitting; the transmit guard handles this.
3. H2C buffers of 16 MiB do not complete on the bench.
4. The PL design files (XSA, bitstream, hwh) are not included. HF lists them.
5. Host-side verification uses emulated hardware. Bench acceptance follows the ATP.